Behavior Management

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Mission: To advance the science and art of health through education, service, scholarship and social accountability

Quiz Time!

Today's Objectives

- Understand the types and how to use various behavior management techniques
- Be able to provide examples and context for correct usage of different behavior management techniques

Recap: Local Anesthesia

- #1 cause of failed behavior management is pain
- Regrettably, local anesthesia administration is the **most challenging** step in pediatric treatment
  - #2 is placing the rubber dam (or other isolation)

Mom: The dentist isn't so bad
The kid in the other room:

Behavior Management

What is behavior management?

- Hallmark of pediatric dentistry: the ability to successfully complete needed dental care
- Utilizes both communicative and non-communicative techniques

Non-pharmacologic behavior guidance

Euphemisms

- Disguising terms/procedures in a way a child might understand the procedure without using ‘scary’ language.
- Examples:
  - A radiograph is a ‘picture’ not an ‘x-ray’.
  - Fluoride varnish is ‘vitamins’
  - Saliva ejector – “straw”
  - Filling – “sticker”
  - Topical anesthetic – “jelly”
- What might you say for a dental dam + clamp?

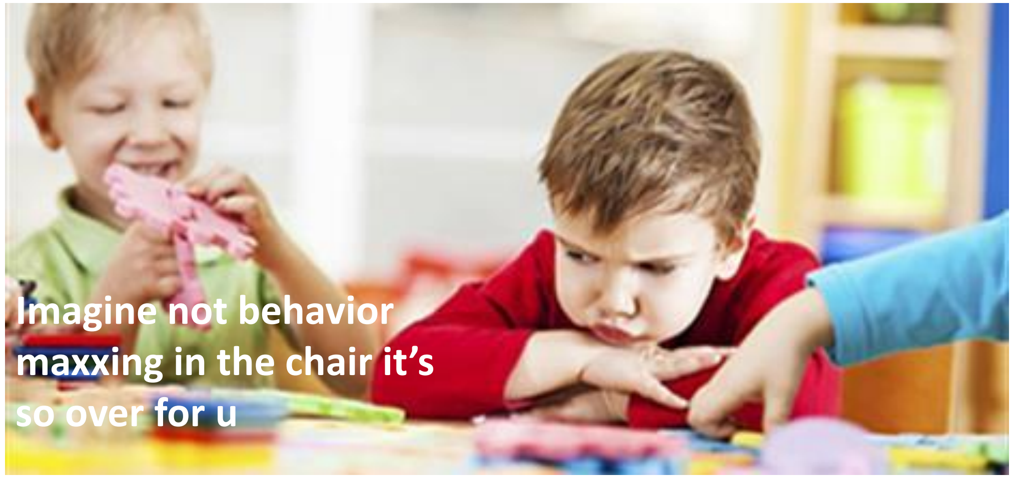
Desensitization

- **Habituation and exposure** (frequent easy visits)
- Parents can simulate at home
- Media can help (first visit videos, kids' cartoons)
- Sibling demonstration (Modeling)
- This technique works particularly well for children with Autism Spectrum Disorder

I cannot stand CoComelon.

Modeling

- Type of desensitization where child watches their sibling/parent (with better behavior) participate in the care first
- Fairly effective, but assumes they are actually watching
- The opposite can also happen where poor behavior is modeled



Distraction

- Divert the patient's attention from what may be perceived as an unpleasant procedure.
- Can be achieved via imagination, clinic design, audio (music/headphones), visual (televisions)
- Giving breaks during a long/stressful procedure would be considered part of distraction techniques.
- Eg: Asking questions throughout local anesthesia

Side note on distraction and desensitization

- Habits at home can influence the success rate of these techniques
  - Children can also be desensitized to the attention-grabbing nature of TV
- Can try to utilize kids' favorite media to improve behavior management

It really be ya own…

Tell-Show-Do

- **Tell, show, do**
- [Tell] Use verbal explanations appropriate for the developmental level of the patient (use euphemisms!)
- [Show] Demonstrate for the patient in a defined, non-threatening way
- [Do] Completion of the procedure
- Kids are way smarter than you think!

TSD Example

Let’s talk fluoride varnish applications.
First, what are the steps?

- Dry the teeth w/ gauze
- Apply fluoride varnish onto smooth surfaces

Convert the steps into euphemisms.

- Dry the teeth w/ a “cotton towel”
- Apply “tooth vitamins” onto the teeth

Show the application.

- Show that the brush is not threatening
- Let them even feel the brush!

Do the application.

Positive Reinforcement/Descriptive Praise

- Give appropriate feedback to promote desirable behavior
- Positive reinforcement rewards desired behaviors, strengthening the likelihood of recurrence
- Can be in many forms: prizes, praise, positive expressions and clapping
- **Descriptive praise** effective – emphasize **specific** behaviors!
- Eg: "Wow! You're such a good helper; opening wide so we can see! Thank you for being so brave and listening to us today."

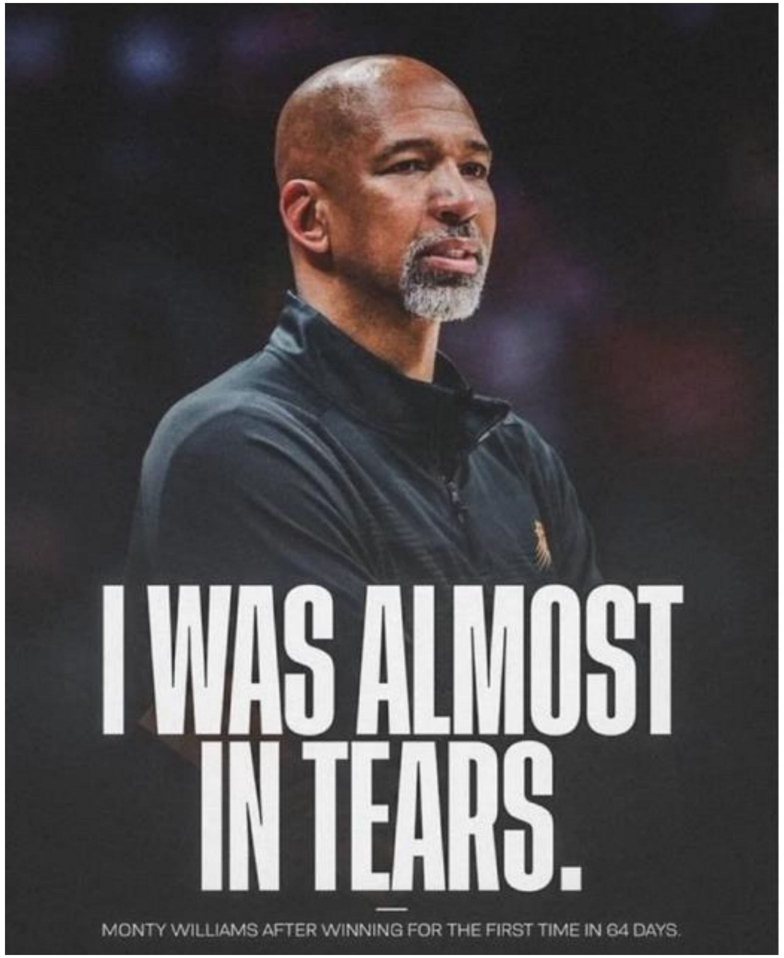
"You were so brave for your prophy today, haha…"

Parents in the room

- Modern parenting styles
- Can be beneficial, can also be detrimental
  - Depends on parent's experiences/projection
- Ask prior to op visits if parents would allow you to explain and discuss procedures

Remember…

- Sometimes things work out, sometimes they don't
- Behavior management doesn't mean it goes perfectly every time
- "Clinically acceptable" and being gracious goes a long way



Papoose (Protective Stabilization)

- Every pediatric dentist will have used this at some point
- Effective for emergency procedures, special needs patients, sometimes sedation
- Requires consent and witness!
- Alternatives include parental restraint or assistant restraint

Pharmacologic management

Why pharmacologic management?

- Applies both to fear and apprehension
- Procedure itself may be challenging (full root ext)
- Medical conditions that benefit from pharma intervention (asthma, seizures)

Pharmacology Problem: Desaturation

- Desaturation is defined as a **5% spO2 decrease from measured baseline.**
  - Eg: 99% spO2 at baseline, 94% would be considered a desaturation event.
- Children desaturate RAPIDLY****
  - Main cause of life-threatening pediatric emergencies is respiratory failure!
- BP: 90-110 / 45-70
- HR: 80-120
- RR: 20-30 breaths/min

Nitrous Oxide (N2O)

What is Nitrous Oxide (N2O)?

- Colorless gas with a very slight, subtle sweet smell
- Effective analgesic/anxiolytic triggering CNS depression and euphoria with little effect on the respiratory system
- Initiated by activation of opioid receptors and **GABA** (gamma-aminobutyric acid) **receptors**
- Absorbed rapidly from alveoli into bloodstream; onset within 2-3 minutes

N2O Cont'd

- **Typical max allowed concentration is 50% N2O / 50% O2.**
- Most people find **the 'sweet spot' at 30-40% concentration.**
- Children respond better to N2O than adults.
- Start w/ 100% O2 for 1-2 minutes. Titrate via 10% intervals.
- N2O **potentiates other sedatives** and should **not** be used in conjunction with other medications unless proper training has been done.

Indications for N2O

- Fearful, anxious patients
  - Asthma is included in this category! (esp stress-induced)
- Patients with certain special health care needs
- Patient with a **severe gag reflex**
- Patient for whom profound local anesthesia cannot be achieved
- **Cooperative** child undergoing **lengthy** dental procedure

Contraindications for N2O

- Recent upper respiratory illnesses
- Chronic Obstructive Pulmonary Disease (COPD)
- Acute otitis media (ear infection)
- Recent acute head injury
- First trimester pregnancy
- Cobalamin (B12) deficiency
- Methylenetetrahydrofolate reductase (MTHFR) deficiency
- ***Lack of legal consent**
- **Medical consult suggesting otherwise**

Common Side Effects N2O

- Note: In the dental office setting you will **never exceed 50% concentration of N2O**, so these are side effects you will run into at that level/below.
- Most common side effects are **nausea and vomiting**
  - Recommend eating a light meal 2hrs prior to treatment
  - Seems to affect teenagers and adults more than younger children (which is why we do gradual titration!)
- Diffusion hypoxia
- Side Note: N2O gets less effective with each consecutive use.

Diffusion Hypoxia

- N_2O has rapid uptake in alveoli. O_2 and CO_2 in the alveoli are diluted by the gas and less O_2 is present, causing desaturation.
- AGAIN: *Children desaturate faster than adults*
- Always administer 5 minutes of 100% O_2!

Oral Conscious Sedation

- Considered moderate sedation
- Uses pharmacologic medicines such as:
  - Benzodiazepines (Diazepam, midazolam, etc)
  - Narcotics (Meperidine)
  - Sedative-Hypnotics (Chloral Hydrate)
- State-based regulations vary
- DO NOT USE THESE MEDICATIONS ON CHILDREN WITHOUT FORMAL TRAINING!

IV / General Anesthesia

- Either a dental anesthesiologist or medical anesthesiologist
- Can be done in office or hospital setting depending on appropriate case selection, medical consult, etc.
- Used for extensive decay, poor behavior, complex medical needs or sometimes all three!
  - Most kids can only tolerate two operative appointments